

The Food Composition Triumvirate: Ingredients, their Characteristics and Processes

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Abstract

Recently a Minimum Information about Food Composition sample metadata controlled vocabulary was launched by the USDA. To support the internationalization of this specification, curators from the FoodOn food ontology have been focusing on a descriptive triumvirate of food process terms, related food materials, and the characteristics of those materials relevant to a given process such as cooking, or food forming. We show how food processes often have related named output materials and adjectives, each of which is helpful for an ontology to provide terms for, especially as they are found in both analytic laboratory, industrial and home recipe making contexts. For example, a (verb) “baking process” produces a (noun) “baked good” material entity, which has a “baked” (adjective) characteristic. This triumvirate of ontology terms is managed in a template table where the relationships between process model parts and descriptors is evident. The table covers the current set of MIFC terms as well as many other more fine-grained process distinctions. The curation illustrates how several communities - nutritional, public health, agricultural, culinary, academic, and industrial research - can meet to develop standards within an interwoven life-science ontology paradigm.

Keywords

food systems, ontology, food processing, recipe, cooking

1. Introduction

As consumers, pleasurable sensory food characteristics often come to mind as we make decisions about acquiring, preparing and eating food - for example the freshness of vegetables or meat, fruit ripeness, and kinds of cuisine. Sensory quality is often considered the most important factor in explaining food choices (Giacalone et al., 2022), but other pragmatic or emotional motivations for eating exist, such as the social environment and consumption context (family or group company; establishment; cultural event), preparation process and convenience (availability and substitution of ingredients; preparation skill and time; scalability and expense; food storage capability), or familiarity, novelty, emotion and memory associated with food. Additionally, consumers are being targeted by social-media and commercial personal health messaging about selecting nutritious food and a varied diet, and fostering a healthy, bioactive gut microbiome, all towards supporting physical activity and lifestyle ambitions.

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The benefits of achieving clarity and consistent food and food component language usage have been advocated by many researchers; a few examples suffice here. A more accurate recasting of dietary recommendations based on nutritional research findings that point to “reduced risk factors for cardiovascular disease, hypertension, diabetes, overweight or obesity, and colorectal and prostate cancers” requires simple moves like resolving the inconsistent use of “legume” vs “pulse” and related characteristics arising from processes like drying, or milling into flour that “make it difficult to compare and combine studies” and assess dietary intakes (Mitchell et al., 2022). Sufficient nutritional consumables assessed from a food composition perspective is not the only factor involved in optimising health. Bioavailability of nutrients and energy depends on optimizing the “food matrix” - the context in which food can be utilized by the body during ingestion (Schönfeldt et al., 2016). Research in this area requires a vocabulary about food processing, both in preparation and during and after ingestion. Food characteristics such as its liquid, semi-solid and solid form, and ensuing shape, texture, and contextual distinctions such as size, thickness, hardness, viscosity, and lubrication “influence oral processing, eating rate, and intake.” (Forde & Bolhuis, 2022). Food safety also bears on individual and population health, requiring a parallel analysis of how pathogenic organisms might propagate through the full life cycle of food production. A food sample foodomics (i.e., metabolomics, ionomics, etc.), genomics and metadata analyses seeks to reveal where contamination risk arises within time and temperature sensitive thermal and other processing treatment of food, including recipe making. (Griffiths et al., 2017).

Before getting to the food characteristic triumvirate, we should acknowledge that research tools must grapple with many food system contextual levels that food samples exist within. Most broadly dataset comparison requires the observational or experimental metadata involved in entities being measured. Research data management projects like ISA-Tab (González-Beltrán et al., 2014) and the Experimental Design Assistant (Percie du Sert et al., 2017) seek to describe experimental design to a fair degree but depend on other frameworks for the vocabulary to describe the focus of study. Observational ontologies, for example, the Extensible Observation Ontology (OBOE), or parts of the Semantic Sensor Network Ontology (SOSA) (Janowicz et al., 2019; Madin et al., 2007) focus on observation components, with for example a SOSA “hasFeatureOfInterest” relation to point to the material (Feature-OfInterest) the observation is about, but also do not themselves define a framework for the kinds of observable material or its characteristics, known as observation semantics, which are obtained from a separate resource (Villa et al., 2017). The Interoperable Descriptions of Observable Property Terminology (I-Adopt) goes a bit further in trying to express some semantic context of a variable being measured. It uses relations such as “hasMatrix” or “hasContextObject” to describe an immediate environmental context in which the material existed, and a “hasConstraint” relation to point to descriptors of the material (Magagna et al., n.d.). I-Adopt’s SKOS representation fulfills an archival and information retrieval objective, but its “context semantics” language is limited, serving as a pragmatic substitute for referencing a natural or anthropological material within the context of a mereotopological model, and is done in many cases where there are no existing standard models, or because it is not economical or convenient to do so. Similarly, The Process and Observation Ontology (PO²) utilizes SOSA as a generic framework for describing observations and the I-Adopt variable design pattern to complement the description of the entity being observed (Dervaux et al., 2021). PO² reuses SOSA concepts such as ‘Procedure’ and ‘System’ to describe observational materials and methods, setting the generic context of the experiment. It also uses I-Adopt concepts such as ‘Entity’ and ‘Variable’ to allow for adding complementary context objects at the variable level which also serve to constrain the context of the property being observed.

To finally see a subject material modeled in its own right requires an integrated framework of domain-specific views, which is accomplished by 3 components. First, a categorical separation is required for material entities, their characteristics, context and parthood, roles they play, and processes they are involved in. This often comes in the form of a top level ontology (TLO). For a TLO we use the Basic Formal Ontology (BFO) (Arp et al., 2015).

Second, underlying ontologies are engineered to specialize in more domain specific semantic views according to the TLO distinctions. We have found that food systems research in agriculture and health

fits within the BFO-compatible OBO Foundry family of natural science ontologies, which cover chemistry, biology, environment, health and disease domains (Jackson et al., 2021). FoodOn is a mid-level OBO Foundry ontology dedicated to describing non-branded food and its processing in the entire food production, distribution and consumption arc (D. Dooley et al., 2024; D. M. Dooley et al., 2018). In OBO Foundry many material entity characteristics (like weight, length, and colour) are named within the Phenotype ontology (PATO), a biology-focused ontology (Gkoutos et al., 2018). In this paper we focus on FoodOn characteristic development with an intention to migrate FoodOn’s general material characteristics to PATO where appropriate.

Third, a shared and reusable set of mereotopological relations is required. Only by knowing the modeled, relation-linked context of an entity and its characteristics do we gain the machine readable fidelity to determine if two or more datasets are comparable on a given variable. OBO Foundry’s Relation Ontology (RO) supplies a collection of mainly object property terms including “**has characteristic**” which attaches an entity, including processes, to a variety of characteristic kinds - including dispositions, functions, roles and phenotypes. (RO’s “**has quality**” object property is more limited because its definition mentions its domain is restricted to independent continuants, which excludes processes.)

The reader may encounter OBO Foundry’s recently introduced BFO-compatible Common Ontology for Biology and Biomedicine (COB) for general use as a simplified, shared top-level entry point for subordinate ontologies to organize categories under. COB reuses PATO’s “characteristic” term, which substitutes broadly for BFO’s “quality” term, defined as a specifically dependent continuant (SDC), meaning that if an instance of its bearer ceases to exist, its instance of a quality ceases as well. PATO’s general food-related characteristic terms are referenced in Figure 1. Here we also allow that a characteristic may involve not just immediately perceivable descriptors such as color, but also spatiotemporal information, namely characteristics like “age” (measured say as 9 years, or categorical “old”) or “irradiated” which depend on information about events in the past.

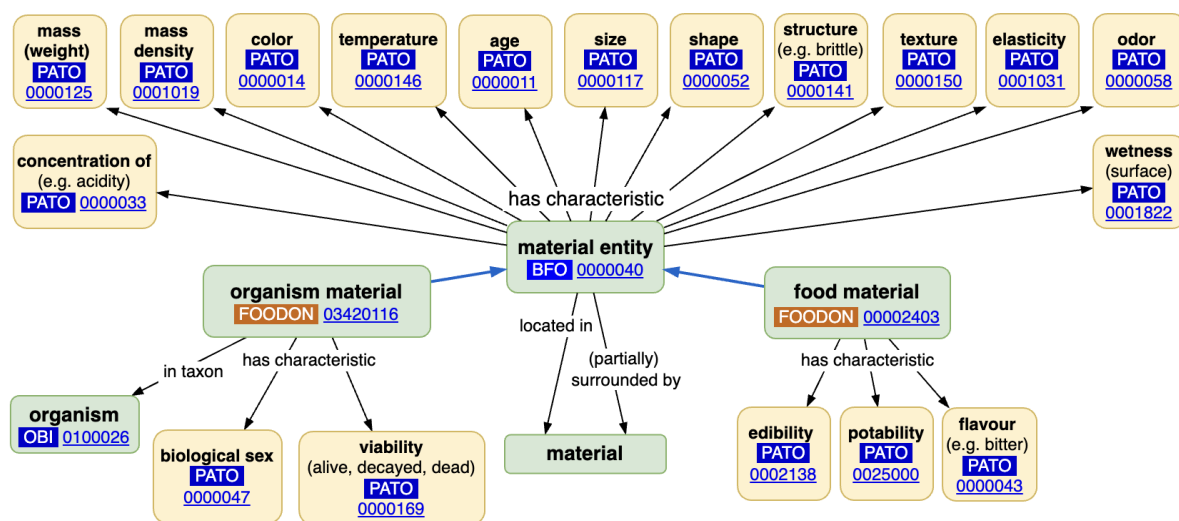


Figure 1: Food materials and organism materials are subclasses of a general material entity class, and each of these kinds inherently has characteristics and mereotopological relations. Food material has a “food role” which is suggested when certain kinds of characteristics are applied to a material entity.

1.1. A process-material-characteristic triumvirate approach

In support of previous work on food process modelling (D. Dooley et al., 2022), an emerging focus for FoodOn is to cover the linguistic triumvirate of descriptive terms that biological or planned material processing modelling involves: a **process** (verb), such as “baking”; the **material entities** (nouns) which a process creates, such as “baked goods”; and the **characteristics** (adjectives) of those goods, such as “baked”. Description of recipes and other process models often shift between reference to all three kinds of term, ideally carrying logical inferences along the way. Our current goal is to create

classes for triumvirate terms; future work will axiomatize the relationships between them explicitly. Often in offering a controlled vocabulary, agencies don't define characteristics per se, but instead offer a description of materials named by a characteristic or process, for example EFSA FoodEx2 defines "raw food" as: "The raw food has not been subject to treatments, or has been subject to such a treatment that the domain specific characteristics are expected to be unchanged." (European Food Safety Authority (EFSA), 2015). Defining characteristics separately allows us to attach them to some entity, thus allowing a reasoner to categorise that entity into appropriate predefined characteristic-driven equivalency classes which provide ontological value.

The OBO Ontology curation community uses "has characteristic" and "is specified input of" or "is specified output of" object properties to express statements such as "baked good" 'has characteristic' some baked", and "baked good" 'is specified output of' some baking". These axioms then guide instance-level graph representation validation of (and inferences about) particular recipes or other processes. Note that while the "specified" input and output relations above apply to planned processes which humans have a hand in, RO also offers short label "input of" and "output of" relations which apply to physical/biological processes that don't involve the intentionality of an agent.

As another example of the triumvirate language, a freezing process (verb) outputs a frozen material (adjective + noun) with a temperature scalar fact about that material (Fig 2.a). More often we say the material is "frozen" (Fig 2.c), which is short for saying it has a temperature characteristic with a "frozen" (i.e. below 0) categorical measure. "frozen" differs from a bearer's "freezing temperature" in that the latter expresses a threshold at which a (liquid) freezing process begins to occur. It is helpful when naming characteristic, process and food material classes to include these senses in the label to dissuade misuse of terms as they appear in ontology search portals.

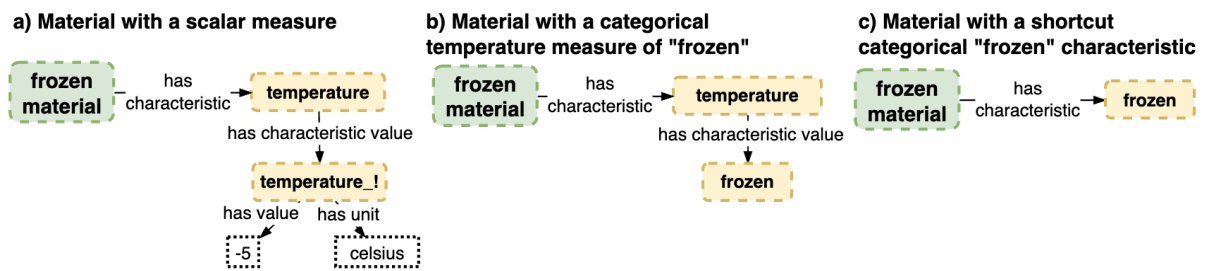


Figure 2: Scalar quantities and categorical descriptors. All dashed boxes are instances of their label class. Note that the "has characteristic value" relation is a tentative proposal for RO adoption.

1.2. Precomposed and Post-composed terms

Raw food materials are a precursor to many basic kinds of processes - drying, freezing, cooking, leading to dried, frozen, or cooked food entities. Since a food material may be in any of these or other states, FoodOn offers an abstract food material parent class consisting only of an "organism material" which has axiom 'in taxon' some [taxon]' (any piece of an organism can be subject of 'in taxon'). Under this term, organism anatomical or more arbitrary butchery pieces, or body products like milk, honey, and maple syrup, are included per organism grouping, species, breed or cultivar. FoodOn has a policy of **precomposing** - i.e. providing an ontology term entry including identifier, definition, and axioms - for any plant, animal, or fungi material used as food, as well as its basic "raw", "dried", "frozen" or "cooked" characteristic state variations, for convenient reference in databases. The precomposed terms are driven by a template system that includes a list of these organisms and a menu system (which includes part terms like leaf, stalk, and fruit) which curators use for selecting which precomposed terms will be generated for an organism.

Precomposing terms for every potential use is a combination bomb (for example, in precomposing a term for every way in which each kind of meat can be cooked, or including prerequisite and normally implicit processing step order as in "juice (condensed, reconstituted)", so FoodOn has allowed only the handful of basic characteristics above in its templating system, beyond these, it is up to FoodOn and other ontology curators to manually provide precomposed terms as needed (for example "halibut steak (breaded, frozen)"). Another example is "chicken hen" which has a logical equivalency to "chicken

and hen”, and so is recognizable by the characteristics of both hen and chicken classes, which respectively involve UBERON “female animal” as a parent class, and an NCBITaxon “‘in taxon’ some ‘Gallus gallus’” axiom. Applications can conveniently reference FoodOn precomposed term hierarchies of organism material and food product in this “food catalogue” way, without having to separately describe a food material’s plant or animal taxon, anatomical or butchery part, or raw state, etc. An alternate approach is to create equivalent name-less “**post-composed**” axiom expressions (like noun phrases or sentences) that entail the same thing from a logical reasoner’s perspective. One ontology may have a class “adzuki bean (dried)” axiomatized as (“seed and ‘has characteristic’ dried and ‘in taxon’ some ‘Vigna angularis’”); this will logically match any knowledge graph instances which has those same axiom statements. An entity can be described by a mixture of both precomposed and post-composed terms.

1.3. The Minimum Information about Food Composition focus

This paper is devoted to detailing FoodOn support of food composition sample term characteristics which are the focus of an emerging Minimum Information about Food Composition (MIFC) standard initiated by the USDA FoodData Central (FDC) project to harmonize food composition datasets (Bloomberg, n.d.; Blumberg et al., 2025). The emerging MIFC standard aims to provide a standardized data structure for the description of food composition data, including enumerated lists of standardized terminology related to foods, their nutritional content and description of the provenance of how samples were obtained and processed (e.g. if a food was obtained from a supermarket or farm). The MIFC standard includes a variety of required, recommended and optional metadata related to the detailed descriptions of foods and nutritional components analyzed from foods. MIFC attributes cover many relevant processes such as food processing and cooking, sample storage, component descriptions and methodology.

Initial development of the MIFC standards arose primarily from the need to standardize the description and representation of data from both the USDA Standard Reference (SR) Legacy database as well as the more recent Foundation Foods dataset. Together these USDA datasets cover more than 4,000 single ingredient food samples (in addition to many multi-component ones) which have been analytically tested or estimated (via nutrient retention factors) for nutritional content.

Here we describe the FoodOn curation work done to ensure that all food characteristics currently included in the MIFC standard are represented by FoodOn terminology, and extend this with additional semantics to connect the material bearing these characteristics, as well as the process responsible for producing them. FoodOn will have many other food-related materials, processes and characteristic terms to be revised as a triumvirate but this work is the vanguard. Describing characteristics for single-ingredient foods is seen as the priority and essential building block that makes estimations of multi-component foods possible. MIFC also includes contextual metadata fields, such as **FoodAcquisitionLocationType** for specifying the kind of food system operator a food sample was obtained from - which can be supplied by a mixture of Industrial Ontology Foundry, ENVO, and FoodOn terms.

2. Methodology

As a foundational contribution to the MIFC project, weekly FoodOn curation group meetings were initiated in 2026 to discuss FoodOn treatment of MIFC terms. Members include people working on the INRAE TransformOn (Weber et al., 2023) and Periodic Table of Food (Jarvis et al., 2024) projects. The group specifically focused on the MIFC categorical picklists for FoodPreservationState, FoodStorageTemperatureState, FoodCookingMethod, FoodForm, FoodRipenessState, and FoodAcquisitionLocationType. FoodOn organizes most of these terms in a single table which broadly covers food processes and related triumvirate terms (The FoodOn Curation Team, 2026). Our paper focuses on MIFC relevant extracts taken from this table, organized in sections below, where each row contains a term with a FoodOn or other ontology term identifier, the parent of that term, the term label and its MIFC label

if any, definition, and with rows color-coded by term type: process terms are blue, materials are green, and characteristics are tan.

Where a full triumvirate of material, characteristic, and process terms is possible, a process's natural language definition ideally describes the transformation leading to an output material; a characteristic is then defined as characteristic of a material that is output from the process; and finally a material is defined as qualified by the characteristic. Many of the process definitions however are difficult to axiomatize, requiring further detailed physical, chemical or perceptual modelling to survive the aristotelean genus-differentiae category distinctions that axioms are supposed to achieve. Further modelling of terms for specific multi-stage recipes such as for making chocolate will require specialized knowledge (Barišić et al., 2019).

3. MIFC harvested material: raw and fresh

Unlike many food characteristics, terms like raw and fresh have provisos about what hasn't been done with a material rather than what processes it has undergone, and consequently their definitions don't involve a simple "characteristic of a material which is output by a process X" pattern. Instead, FoodOn defines these terms with reference to the composition of food material at point of harvest, which allows for functional flexibility that both natural language and more analytic usage accept (Table 1).

Table 1
Point-of-harvest related terms

Parent term	FoodOn Term	MIFC	Definition
organismal quality	raw	☒	A characteristic of organism material which has a biochemical composition that closely resembles its harvested or subsequent biologically aged state.
organismal quality	raw (not intact)		A characteristic of organism material which has undergone comminution or meat tenderizing, but which is otherwise unprocessed.
food material by characteristic	food (raw)		Food material which is raw.
raw	fresh	☒	A raw characteristic of organism material which has not undergone significant deterioration or spoilage, and which has not been subjected to drying or freezing.
food (raw)	food (fresh)		Raw food which is fresh.

These definitions allow for cleaning, cutting, hulling, storing or trimming after harvest as long as the biological state of component(s) at harvest are maintained. FoodOn's definition of "**raw**" is delicately crafted to capture both the common language and food safety/inspection sense of "raw". It covers food that is still in harvested state, whether that is fresh, overripe or unripe, or naturally dried on the stem (for example pulses like kidney beans), but also food which has been naturally aged, such as fruit in cold storage warehouses, or food which has been cut, acknowledging that from a scientific perspective the nutrient profiles may change due to aging. FoodOn's "raw" is similar to the FoodEx2 noun mentioned above, as well as the National Agricultural Library definition of raw food, "Foods that have not been preserved or processed in any way other than washing, cutting and/or chilling" (<https://lod.nal.usda.gov/nalt/en/page/421>). Note that while a material as a whole might not be raw, some part of it can be, as with steak done rare, or blanched tomatoes. "Fresh", a subclass of raw, involves the non-existence of deterioration, freezing or drying. General materials having these characteristics are "food (raw)" and "food (fresh)". The antonym for fresh, PATO's "decayed" term, may also be useful. The

MIFC **FoodPreservationState** begins with raw and fresh terms, and continues through subsequent dried, aged, temperature and chemical treatments described below.

4. MIFC: dried

The way materials are dried - naturally, through sun exposure, air evaporation, or both, or through artificial methods - can affect nutrition and energy availability, food security and logistics, and ultimately food quality and safety. The MIFC “dried” characteristic is straightforwardly defined as a characteristic of a material which has been subjected to a drying process; it then falls upon the process definition of “drying” to constrain what kinds of materials are involved and the degree of transformation. A new FoodOn “drying” process is noncommittal about the kind of liquid removed or the extent of liquid removal. Subclasses of drying then develop terms covering more particular drying target, technique and extent, ranging from point of harvest context where a pulse may have naturally dried, to artificial drying occurring after harvest. FoodOn matches all of MIFC’s subordinate drying characteristic terms with processes, and also has more specific terms used in research and industry. A food material can have component concentrations of liquids besides water, including ethanol and oil, hence terms like “moist” can cover a sensory effect produced by such liquids. These characteristics can be with respect to surface (PATO “wet” or “dry”) or interior moisture content (“moist”, “dried”). Dehydration or rehydration processes lead to “dehydrated” or “rehydrated” characteristics, both of which implicitly reference a past sensory state. The cutoff moisture content between when a material is considered dried versus dehydrated is left underspecified with the expectation that subclasses based on temperature or moisture limits will be created for industrial or regulatory use cases (sweet or spicy foods are often allowed greater moisture percentages for example). From food safety, nutrition and other perspectives a dried food might be considered raw.

Table 2

A sample of FoodOn drying process related terms.

material component separation	drying		A material component separation process in which water or another solvent is removed from a material.
drying	drying (water reduction)		A drying process in which water is removed from a material.
food material by process	food (dried)		Non-liquid food material which has undergone a water reduction drying process.
physical object quality	dried	☒	A physical object quality of material that has undergone a drying process.
food (dried)	food (dried raw)		Dried food which was dried when in a raw state.
dried	dried raw		A dried characteristic of organism material that was raw when subjected to a drying process.
	dry		A wetness quality inhering in a bearer by virtue of the bearer's not being covered by a liquid .
drying	heat drying process		Drying of material by application of thermal energy.
heat drying process	natural heat drying		Drying by exposure to the sun and/or ambient air.
dried	naturally dried	☒	A dried characteristic of material which indicates that it has undergone natural heat drying.

material component separation	drying		A material component separation process in which water or another solvent is removed from a material.
natural heat drying naturally dried	sun drying sun-dried	☒	Drying by exposure to the sun. A naturally dried characteristic of material which indicates that it has undergone sun drying.
natural heat drying naturally dried	air drying air-dried	☒	Drying by exposure to ambient air. A naturally dried characteristic of material which indicates that it has undergone ambient air drying.
heat drying process dried	artificial heat drying artificially-dried	☒	Drying by the application of artificial heat. A dried characteristic of material which indicates that it has undergone artificial drying.
dehydration	freeze-drying		Dehydration of food material by freezing it within a vacuum to remove moisture via sublimation, which turns ice directly into vapor.
food (dried) artificially-dried	food (freeze-dried) freeze-dried	☒	Dried food which has been freeze-dried. An artificially-dried characteristic of material which indicates that it has undergone freeze-drying.

5. MIFC: food maturation

MIFC covers food maturation, defined generally in FoodOn as the “The biological transformation of food material over a period of time to attain a flavour or food preservation and safety objective.” under **FoodRipenessState** and **FoodPreservationState**. Some food maturation processes, such as food ageing, occur in a storage location with a controlled environment, and some, like fermentation, involve the use of natural or added enzymes, yeasts or bacteria. For plants and fungi, maturation may happen before or after harvest, such as the ripening of fruit on or off the vine, including biological fruits often called vegetables, like tomato, squash, bell peppers, cucumber, and avocado. A distinction exists between non-climactic fruits that only ripen before harvest, such as grapes and berries, versus climactic fruit ripening (such as for tomatoes and bananas), which can happen before or after harvest, and which can be accompanied by artificial ripening methods. FoodOn does not have an elaborate vocabulary for the cheese ripening (affinage) or wine fermentation; these domains are ripe “terroirs” for dedicated ontologies.

Table 3

A sample of FoodOn food maturation terms. FoodOn also includes general terms for ageing durations, wine and cheese maturation, and curing food.

food maturation storage	food ageing		A food maturation process occurring in a storage location with a controlled environment.
food material by characteristic	food (aged)		A characteristic of food that has undergone an ageing process.

food maturation storage	food ageing		A food maturation process occurring in a storage location with a controlled environment.
food age-ing microbial/enzymatic modification	enzymatic food modification		A food ageing process involving the addition of enzymes or the activation of endogenous enzymes.
food maturation enzymatic modification	fruit ripening	☒	A developmental maturation process that has as participant a fruit. Ripening causes changes in one or more characteristics of a fruit (color, aroma, flavor, texture, hardness, cell wall structure) and may make it more attractive to animals and aid in seed dispersal.
food plant physical quality	fruit or vegetable ripeness		A food plant physical characteristic about the ripening stage of development of a fruit (or vegetable).
fruit	unripe fruit		A fruit which is unripe.
fruit or vegetable ripeness	unripe	☒	A fruit (or vegetable) ripeness characteristic in which a fruit has not achieved full maturity and is not slightly ripe.
fruit	slightly ripe fruit		A fruit which is slightly ripe.
fruit or vegetable ripeness	slightly ripe	☒	A fruit (or vegetable) ripeness characteristic in which a fruit has some ripe part or is about to achieve full maturity.
fruit	ripe fruit		A fruit which is ripe.
fruit or vegetable ripeness	ripe	☒	A fruit (or vegetable) ripeness characteristic in which a fruit has achieved full maturity and is ready for utilization.
fruit	overripe fruit		A fruit which is overripe.
fruit or vegetable ripeness	overripe	☒	A fruit (or vegetable) ripeness characteristic in which a fruit has surpassed ripeness and is undergoing decomposition.

6. MIFC: Temperature

Ontologically, a temperature is a characteristic that inheres in a bearer which can range from a material entity that may have cavities and core and surface area regions, such as a cake, a chicken, an ice cube or a room. One can measure temperature according to Celsius or Kelvin scalar variables, or more loosely with terms like hot, warm, cool or cold. A class-subclass arrangement of temperature kinds exposes that underlying classes add constraints about the bearer's physicality, history, or context. For example, a "skin burning temperature" is the characteristic of a bearer relative to some (animal) skin; and a "scalding temperature" is a kind of skin burning temperature that is a function of liquid (oil, water) relative to perceiver. Some characteristics like "chilled fresh", a conjunction of both "chilled" and "fresh", are simultaneously observable, while others combine information which tells us something about a material's history, for example, "fresh frozen" indicates organism material was fresh when initially frozen, while "thawed" (that is, un-frozen) presupposes a previous frozen state. A "chilled" characteristic necessarily references a bearer such as an ingredient (or inside of a refrigerator) as being in a "chilled" state, while "chilling temperature" references the temperature at which some bearer would be considered chilled. Some terms such as "foodsafety chilling temperature" require subclasses pertinent to particular jurisdictions in order to be factual about recommended temperature ranges; meanwhile a "foodsafety chilled" characteristic applies to food within such a range, and "food (foodsafety

chilled)” is food which has been stored for some duration within a temperature range.

Currently MIFC’s **FoodStorageTemperatureState** provides an “uncontrolled” temperature term (collected from an uncontrolled environment) and an “ambient” term which matches FoodOn’s comfortable “room temperature”. This is relative to environments around the globe, consequently an “ISO 1 standard temperature” of 20°C exists to enable formal comparison of food material characteristics, which can be important in recipe making. Declaring categories of temperature with respect to a perceiver is one perspective; the other is declaring temperatures related to processes that control them (Table 4). A distinction is made between general temperature controlled refrigerated, chilled, and frozen characteristics of material, versus foodsafe versions of these which all have more rigid temperature characteristics, sometimes varying by jurisdiction, an important distinction when attempting to model pathogen viability for food safety.

Table 4

A sample of food related temperature characteristics, materials and processes.

thermal energy modulation	material cooling		A thermal energy modulation in which heat is removed from a material or (indirectly) its environment.
material cooling	cooling material to a threshold		A material cooling process which commences when a material is placed into an environment which has a cooler temperature.
cooling material to a threshold	material chilling		A cooling material to a threshold process which is at or above the freezing point of water, and less than a cool (~7°C) temperature.
food material by characteristic	food (chilled)		Food material which is the output of a cooling material to a threshold process.
food (chilled)	food (foodsafe chilled)		Chilled food which is within a recommended foodsafe chilling temperature.
physical object quality	chilled	☒	A chilling temperature characteristic of material indicating that it bears this temperature by having undergone a cooling material to a threshold process.
chilled	chilled fresh		A chilled temperature of material which also holds a fresh characteristic.
chilled	foodsafe chilled	☒	A chilled temperature of material that is a temperature which satisfies safe food storage standards.
cooling material to a threshold food preservation	refrigeration		A cooling material to a threshold process which is carried out during short or long-term food storage.
refrigeration	refrigeration with a refrigerating agent		A process of refrigeration involving a refrigerating agent.
food material by process	food (refrigerated)		Food material which has undergone refrigeration.
food characteristic	refrigerated	☒	A characteristic of food material which has undergone refrigeration and/or is within a refrigerator.
cooling material to a threshold food preservation	material freezing		A cooling material to a threshold which transition its phase from a liquid or gas into a solid.

thermal energy modulation	material cooling		A thermal energy modulation in which heat is removed from a material or (indirectly) its environment.
food material by characteristic temperature	food (frozen)		Food material which has undergone freezing.
	frozen	☒	A quality inhering in a bearer by virtue of the bearer's being kept below its freezing point.
food (frozen)	food (fresh frozen)		Food which has a fresh frozen characteristic.
frozen	fresh frozen		A frozen characteristic which indicates the bearer (organism material) had a fresh or blanched state immediately before being frozen.
frozen	frozen after cooking		A frozen characteristic of food material which was cooked and then frozen.
material freezing	food natural freezing		A material freezing process in which ambient natural environment sub-zero temperatures are used to freeze food material.

7. Cooking and Heating

Cooking requires a delicate definition that encompasses protein denaturation accomplished most often by the application of heat, but also by other methods like acidification (in ceviche for example). Cooking by heat distinguishes between dry and moist/liquid based cooking (for example fried, dry-roasted, boiled, steamed), all of which can be carried out to varying times and extents of internal or external cooked states. Cooking techniques lead to an amalgam of organoleptic characteristics - smell, texture, vision and taste, for example roasting which is scientifically characterised by the Maillard reaction of non-enzymatic browning which occurs in making roasted coffee, meat, vegetables, and chocolate. The forty MIFC **FoodCookingMethod** menu characteristic terms also include a few antipodes like “uncooked” and “unheated” which are related to raw and fresh terms.

The formalization of warming terms reveals implicit knowledge, that for example “thawing” involves heating frozen material to above 0°C but not changing its unfrozen biochemical state beyond that. Recipes assume room temperature ingredients unless specified, hence humans and robots must also factor in transitioning refrigerated food material to room temperature to estimate cooking times.

8. Food Preservation

In addition to a selection of the above raw, fresh, maturation, dried characteristics, and a “heat-treated” characteristic that covers all thermal cooking methods, MIFC’s **FoodPreservationState** menu includes selected chemical, atmospheric (pressure & gas) and radiation preservation characteristic terms: candied, jellied, pasteurized, canned, brined, pickled, kippered, marinated, irradiated, and ultraviolet light exposed. FoodOn backs these up with the triumvirate process and material terms. FoodPreservationState also provides “shelf stable” for food material which from a food safety perspective can be safely stored at room temperature, and “durable shelf life” for food which is shelf stable and maintains its quality and desired palatability, as well as fulfilling any nutritional declarations up to some date of minimum durability.

9. Combining, shaping, agitating, comminution and portioning of material

The remaining food processes of combining, shaping, agitating, portioning and separation of material bring us to general material science transformation terms involving modifying the physical shape, size and often identity of materials over time, which are still being organized in FoodOn and OBO Foundry. The MIFC **FoodForm** menu outlines materials in various forms as a food composition researcher would perceive them, starting with basic material phases: liquid, viscous liquid, semi-solid, and solid. Comminution, the reduction of solid material into smaller particle sizes - for example by cutting, slicing, milling, and grinding - leads to paste, powder, and puree terms. Material combination and agitation outputs range from foam and emulsion mixtures and materials like dough, all with characteristic concentrations. Shaping material involves many characteristics such as round, flat and tubular, as material transformation objectives. Finally, chemical and physical distinctions - whether a liquid is immiscible with respect to another, or whether materials are countable, like apple and orange pieces - can determine other characteristics of material transformation processes such as whether they involve countable objects, or mass noun forms, and are reversible or not.

10. Conclusion

The triumvirate round of food characteristic vocabulary curation described above is aimed firstly at supporting the MIFC standardised vocabulary for food sample composition and contextual description, and more broadly it supports formalizing sample collection and processing protocol vocabulary. FoodOn's direct and tangential characteristic term development will likely lead to more discussion about use-cases and new term requests or adoption in databases such as the USDA's Foundation Foods and the Periodic Table of Food Initiative.

Lastly, the triumvirate will support the upcoming FoodOn Recipe model, where each process step involves ingredient combination, agitation, separation, comminution, and thermal treatments - all ideally accessible in the plain language of home cooking, but also in the formal context of industrial food formulations, and in food safety risk assessment, areas where we can start to share logic-driven algorithms for analysis and prediction. We seek a processing model that is logically linked to kinds of materials and their characteristics in order to bear more reasoning power. Future development for the recipe use-case involves creation of a reference database of ingredients, specifying their characteristics at certain temperatures and pressures, such as the specific heat capacity of ingredients, which for example determine what temperature a mixture of them will have. As well, the role of devices and their characteristics needs to be integrated into the process model, to enable a plug-and-play decisionmaking model that can scale with the inputs and output demands at hand. Ultimately we hope the triumvirate will contribute to research on the phenotypic and genotypic "foodomics" aspects of food on the production side, as well as the consumer-side nutritional impact of food processing and bioavailability.

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Declaration on Generative AI

The author(s) have not employed any Generative AI tools.

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